Suggested Topics for FYP 2026-27

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Topics in 2026-27

- YYC1: An EEG-Based BCI Communication System
  - Quota: 2 students (individual or as one group)
  - Industry Co-advisor: LAURRY AI

- YYC2: An EEG-Based BCI Application
  - Quota: 2 students (individual or as one group)
  - Industry Co-advisor: LAURRY AI

YYC1: An EEG-Based BCI Communication System

Quota: 2 students

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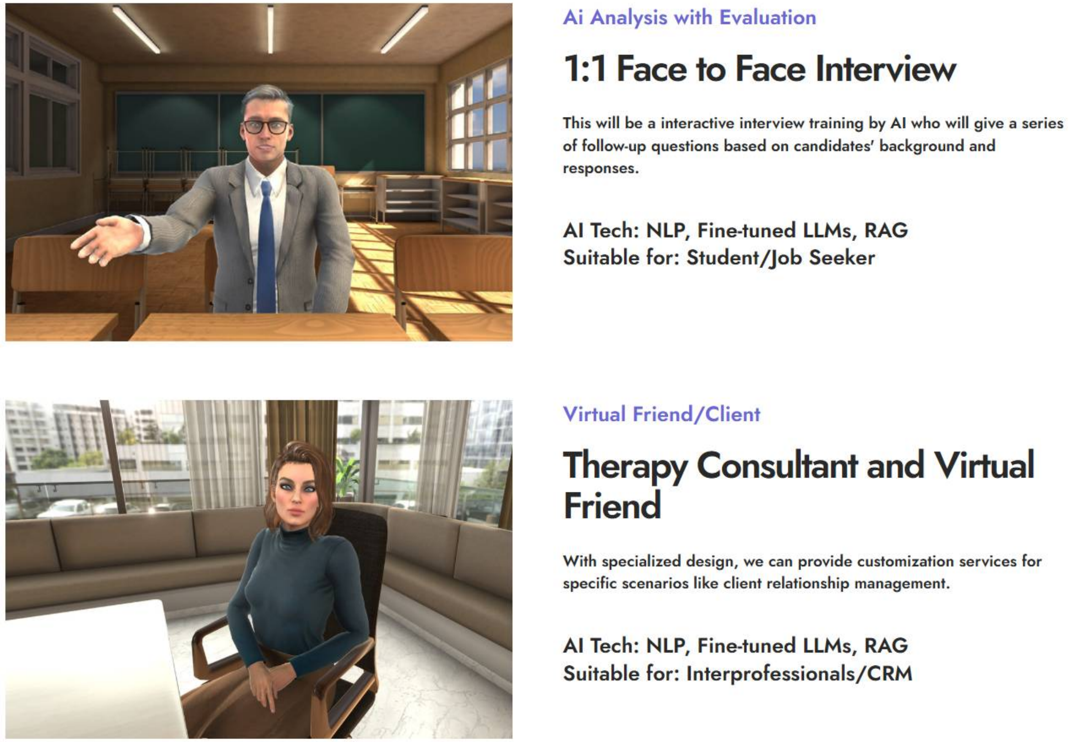
YYC2: An EEG-Based BCI Application

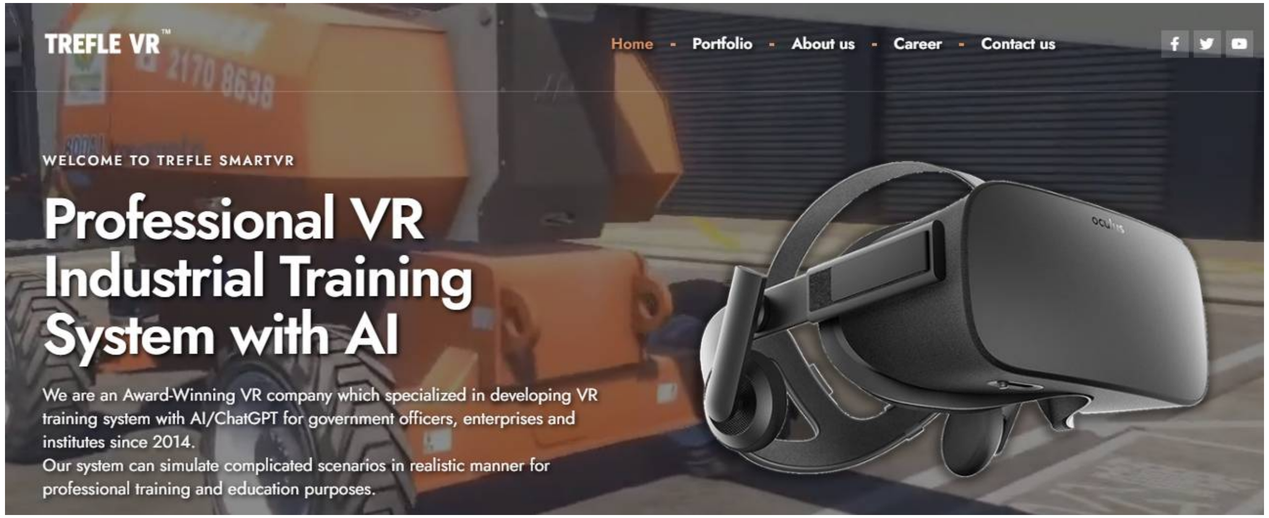
Quota: 2 students

Industry Co-Advisor: LAURRY AI

About the Industry Co-advisor:



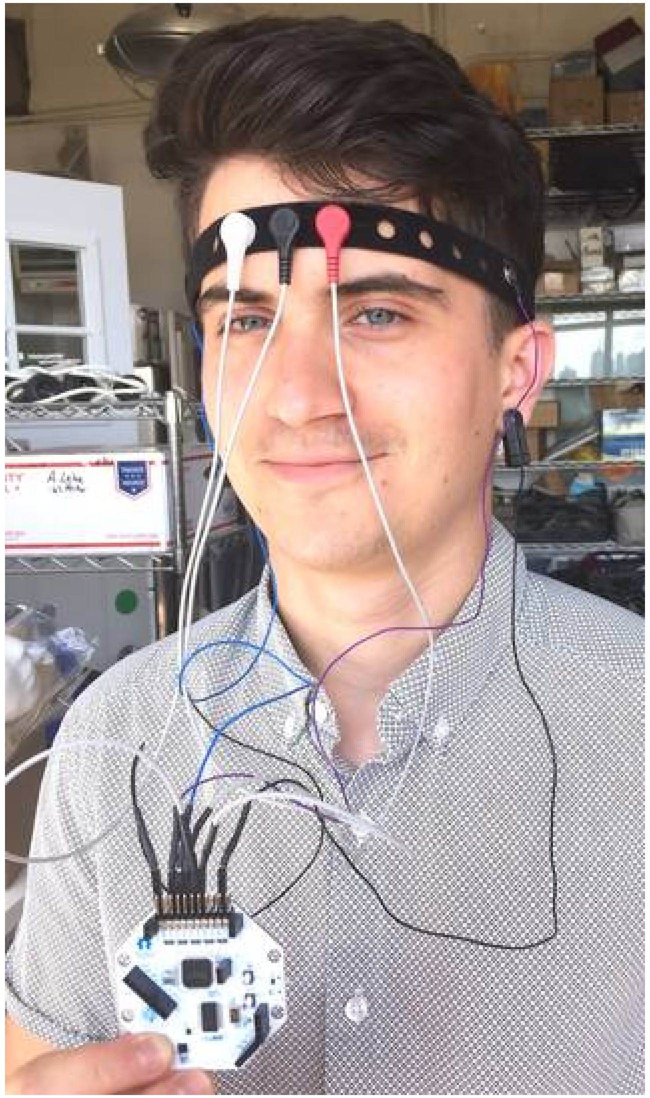
(Source: Laurry.com)



Objective and Overview

- **Objective:** Develop a Brain-Computer Interface (BCI) system using EEG signals for individuals with severe physical limitations to communicate through brain activity.

- **Key Points:**
  - Target users: Individuals with conditions like locked-in syndrome, severe paralysis, or neurodegenerative diseases.
  - Enable communication without speech or physical movement.
  - Real-time decoding of EEG signals to generate words or phrases.

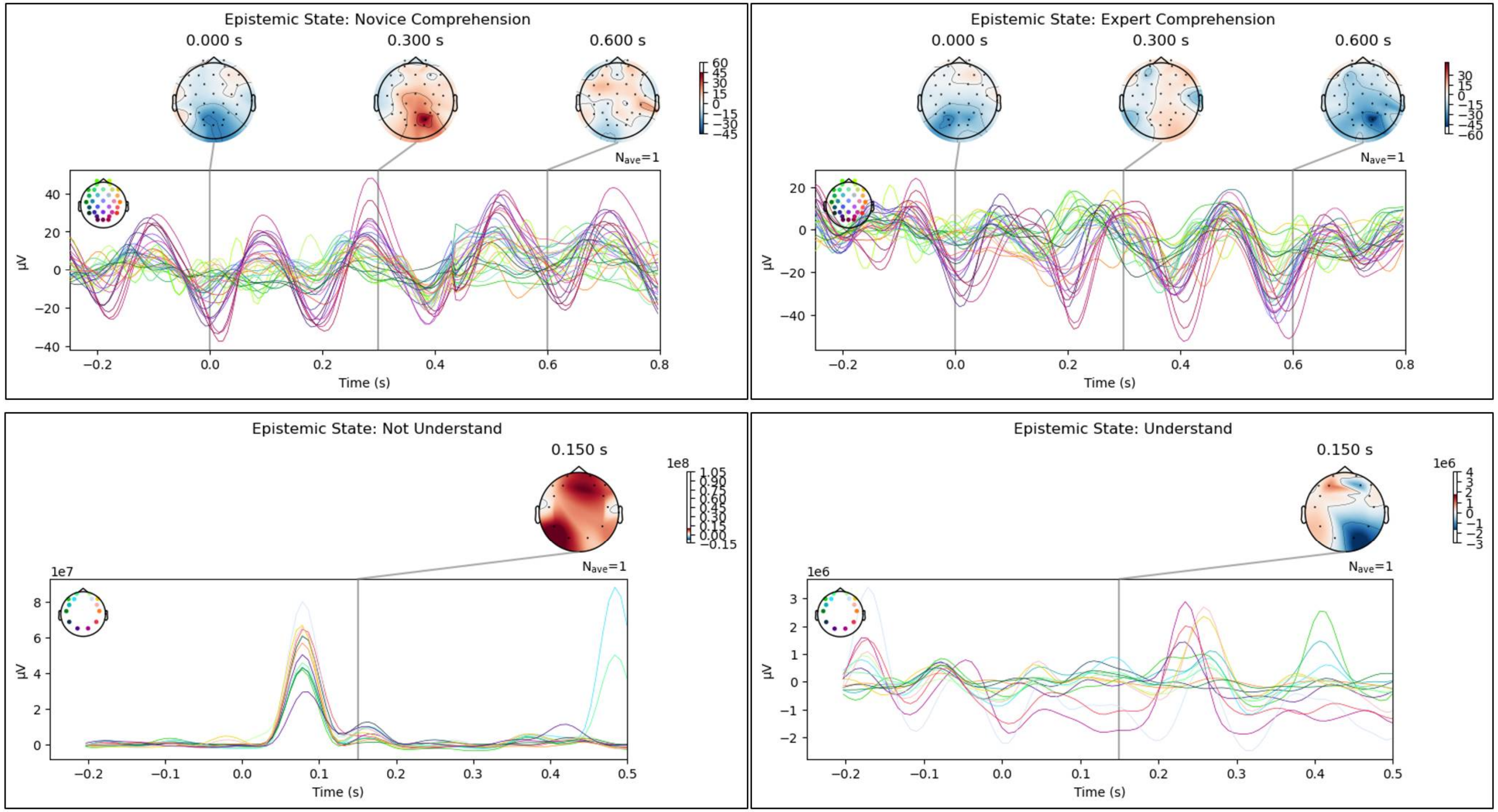


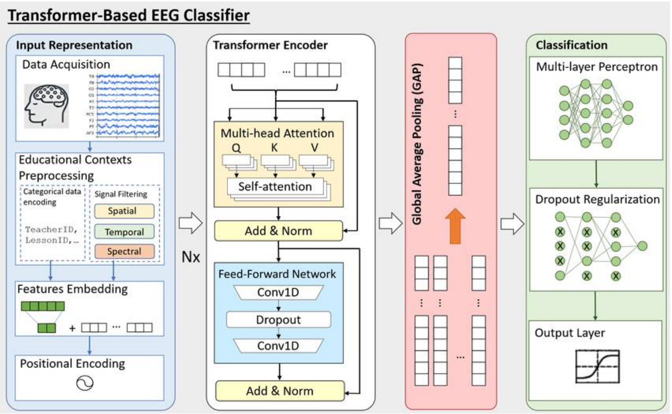
(Source: OpenBCI)

Methodology

- **1. EEG Signal Acquisition**:
  - Use EEG headsets (e.g., OpenBCI).
  - Capture signals related to cognitive states (e.g., focus, intention).
- **2. Signal Processing**:
  - Preprocess EEG signals to remove noise and artifacts.
  - Extract features (e.g., alpha, beta waves) relevant for command classification.
- **3. Machine Learning Classification**:
  - Apply machine learning algorithms (e.g., CNN, Transformers) to classify different cognitive states.
  - Interpret signals into actionable commands (e.g., “yes/no” or selecting letters).
- **4. User Interface Design**:
  - Display interface with words/letters.
  - Users focus on desired letters/words to select them based on EEG signals.

EEG Classification using Transformer Models





Yuen-Yan Chan, Rosanna, "EEG Transformer for Classifying Students' Epistemic Cognition States in Educational Contexts," *IEEE Access*, vol. 13, pp. 23935-23949, 2025.

YYC1: Mobility and Physical Limitations

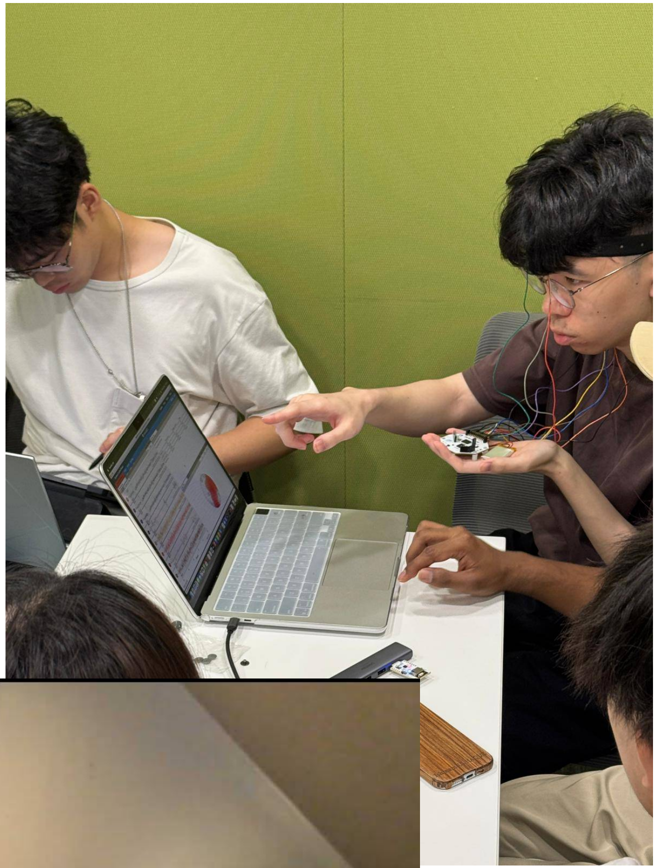
- People with cerebral palsy and/or other conditions have varying degrees of difficulty with movement.

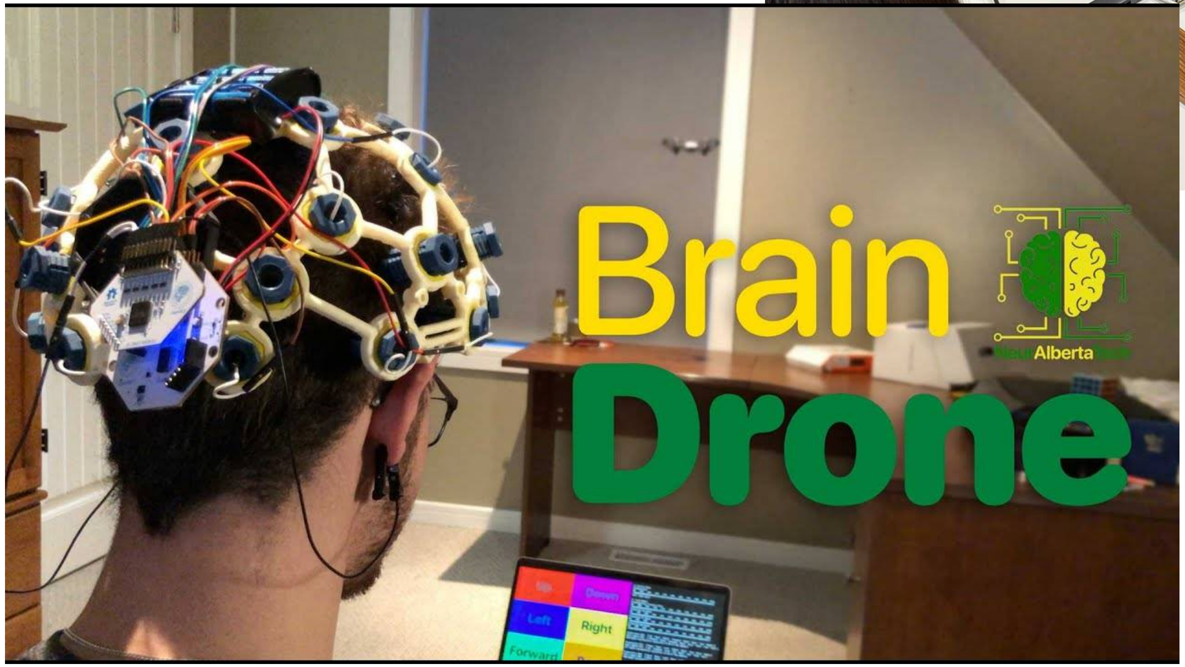
- Some people having dyskinetic CP cannot speak or write.

YYC2: Neuro Applications

- EEG signals can also support the development of non-invasive BCI applications

- Including entertainment games and interactive environments.





Required Skills & Tools

- **EEG Signal Processing** – Understanding EEG data, noise reduction, and feature extraction (e.g., filtering, Fast Fourier Transform, Wavelet Transform).
- **Machine Learning & Deep Learning** – Model training for classifying brain signals (e.g., CNNs, RNNs, Transformers).
- **Brain-Computer Interface (BCI) Development** – Implementing real-time signal interpretation for communication applications.
- **API Integration with LLMs (Optional)** – Enhancing communication by integrating EEG-based intent recognition with language models.

Expected Deliverables

- **Fully Operational System**
  - A working prototype demonstrating the core functionalities of the project.
- **User Interface**
  - Intuitive and user-friendly interface for interaction with the system.
- **Testing & Validation**
  - Evidence of functional and user testing to ensure performance meets the specified requirements.

- **Plus other FYP requirements (logs, proposals, reports)**

Any Questions?

Please email to yychan@ie.cuhk.edu.hk

(Note: Inquiry such as “Please tell me more about the project!” or “Can we have a meeting to learn more?” without describing your skills and background and why you are interested in the project(s) will not be entertained.)